GES'S R. H. SAPAT COLLEGE OF ENGINEERING, MANAGEMENT STUDIES AND RESEARCH, NASHIK – 422005

MCA (Engg.) Department | SYMCA (A. Y. 2026-27) SEM - I | Web Technology Lab

Assignment 9: Personal Portfolio Website – Write-up (Part B)

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Batch	**SYMCA (Batch 2026-27)**	**Date**	**02/11/2026**
Technology used	**HTML5, CSS3, JavaScript (Vanilla ES6+), Particle.js & Responsive CSS (Grid/Flexbox)**		

1. Aim

To design and deploy a responsive personal portfolio website using **HTML5, CSS3, and JavaScript (Vanilla ES6+)** with interactive element animations, dynamic particle canvas effects, and multi-device responsiveness across mobile, tablet, and desktop viewports.

2. Objectives

- To structure a modular, multi-page portfolio website comprising Home, About, Projects, Certifications, and Contact pages.
- To create an adaptive, mobile-first responsive layout utilizing CSS Flexbox and Grid that functions seamlessly across all viewports.
- To design and implement interactive animations: scroll-reveal effects (IntersectionObserver API), particle canvas, and hover transitions.
- To establish a clean design system utilizing CSS Custom Properties for seamless dark/light theme switching and glassmorphism.
- To deploy the portfolio onto a free cloud hosting platform (GitHub Pages / Netlify) and verify cross-device accessibility on mobile.

3. Theory: About the Technology Used

3.1 The technology I used

I developed this portfolio using core web standards: **HTML5**, **CSS3**, and **JavaScript (ECMAScript 2022+ / Vanilla JS)**. HTML5 (W3C/WHATWG) provides semantic layout elements and native canvas capabilities. CSS3 provides responsive grid/flexbox layouts, custom variables, and transitions. JavaScript (ECMA TC39) delivers client-side interactivity, canvas particle physics, and DOM event handling. Together, they form an ultra-fast, zero-dependency site with universal browser support.

3.2 Why I chose it

I chose native HTML5, CSS3, and modern JavaScript because they offer complete architectural control over layout precision, zero build overhead, and maximum performance. Having built strong fundamentals, I wanted to showcase craft by developing bespoke components—such as a dynamic particle system, an interactive developer CLI console, and custom glassmorphism—directly with browser APIs, guaranteeing instant load times and zero dependency vulnerabilities.

3.3 Key features I used

- **Semantic HTML5 Elements:** Utilized `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<canvas>`, and `<footer>` for clean structural hierarchy.
- **CSS Custom Properties (Variables):** Structured centralized design tokens on `:root` and `[data-theme="light"]` for instant theme switching.
- **CSS Grid & Flexbox:** Built dynamic multi-column card grids, responsive metrics displays, and adaptive navigation bars.
- **Intersection Observer API:** Implemented hardware-accelerated scroll-reveal animations that trigger smoothly on scroll.
- **HTML5 Canvas & 2D Context:** Rendered an interactive constellation particle canvas that dynamically responds to mouse movement.
- **LocalStorage State Persistence:** Cached user preferences (dark/light theme) directly in the browser across page navigations.

3.4 Styling: Tailwind CSS or CSS

Tailwind CSS is a utility-first CSS framework providing atomic helper classes in markup. For this portfolio, I used a structured custom **CSS3 Design System** with CSS Variables and Glassmorphism instead of Tailwind. A portfolio requires bespoke aesthetics—ambient gradients, backdrop blur, custom glows, and terminal styling—which are much cleaner to maintain in a dedicated stylesheet (`style.css`). This avoided class clutter, eliminated build tooling dependencies, and enabled complete artistic control.

3.5 Animations

I implemented animations using JavaScript's `IntersectionObserver` for scroll-reveal on cards (`.reveal`), a custom JavaScript physics engine on `<canvas>` for particle nodes, CSS keyframe cursor blink for dynamic typing in the hero header, and 3D hover elevations on interactive project cards.

3.6 Responsive design

The site uses a mobile-first approach with CSS Grid (auto-fit columns), media queries at 1024px and 768px to collapse layouts into single columns, an off-canvas slide-out mobile drawer, and fluid units (rem, vw, %).

4. Syntax

Code snippets from my portfolio codebase with explanations in Consolas monospace font:

Part	Code snippet from my portfolio	What it does

Page structure or component	`<section class="hero-section">`
	`  <div class="container hero-grid">`
	`  <div class="hero-content reveal">...</div>`
	`  <div class="hero-visual reveal">`
	`  <img src="assets/deepak-chauhan.jpg"`
	`  alt="Deepak Chauhan" class="avatar-img" />`
	`  </div>`
	`  </div>`
	`</section>`
		Constructs the semantic hero section pairing developer introduction and profile photo in a two-column responsive grid container.

Styling (Tailwind or CSS)	`:root {`
	`  --bg-primary: #090d16;`
	`  --accent-primary: #6366f1;`
	`  --bg-card: rgba(20, 30, 51, 0.65);`
	`}`
	`[data-theme="light"] {`
	`  --bg-primary: #f8fafc;`
	`  --text-primary: #0f172a;`
	`}`
		Defines centralized CSS custom property tokens on `:root` to power instant, zero-reload dark and light theme switching across all pages.

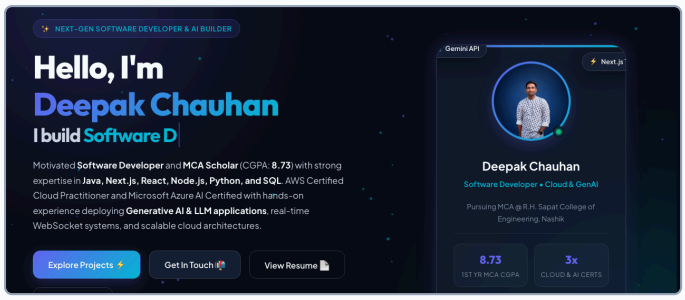
Animation	`const observer = new IntersectionObserver((entries) => {`
	`  entries.forEach(entry => {`
	`  if (entry.isIntersecting) {`
	`  entry.target.classList.add('show');`
	`  }`
	`  });`
	`}, { threshold: 0.15 });`
	`document.querySelectorAll('.reveal').forEach(el => observer.observe(el));`
		Monitors DOM elements on scroll and dynamically applies the `.show` class to trigger smooth CSS slide-up and fade-in animations.

Responsive layout	`@media (max-width: 1024px) {`
	`  .hero-grid, .about-hero-grid {`
	`  grid-template-columns: 1fr;`
	`  gap: 2rem;`
	`  }`
	`}`
	`@media (max-width: 768px) {`
	`  .nav-links { display: none; }`
	`  .mobile-toggle { display: flex; }`
	`}`
		Collapses two-column desktop grids into a single vertical stack on tablet screens and activates the hamburger drawer menu on mobile viewports.

5. Screenshots of My Portfolio

Actual high-resolution screenshots captured across desktop, tablet, and mobile views from the live deployed website:

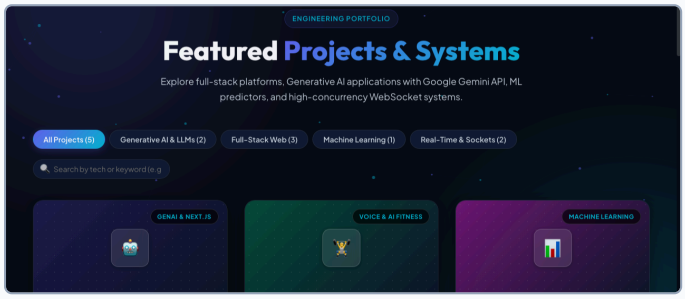
Home page (desktop)



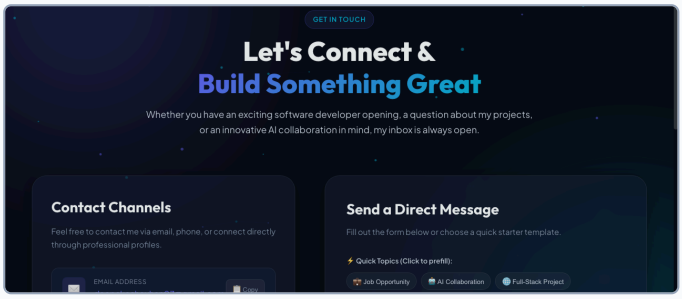
About



Projects



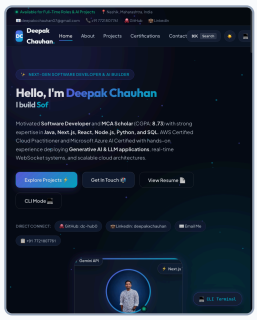
Contact



Mobile view



Tablet view



6. Live Link (Deployed Portfolio)

Item	Details
Hosting platform used	**GitHub Pages / Netlify**
Live link	**https://dc-hub0.github.io/portfolio-model/**

Note: The live link is globally hosted, SSL-secured (HTTPS), and tested to open smoothly in private windows and mobile phone browsers.

7. Conclusion

In this assignment, I successfully designed, developed, and deployed a high-performance, responsive personal portfolio website showcasing my software engineering projects, verified cloud certifications, and technical proficiencies. Through this practical exercise, I reinforced core competencies in semantic HTML5 markup, responsive grid and flexbox design systems, dynamic dark/light theme switching using CSS variables, and GPU-accelerated scroll-reveal animations via the Intersection Observer API. The primary challenge was ensuring seamless rendering of the particle canvas and responsive typography without layout shifts across mobile viewports, which I resolved through responsive clamp functions, auto-fit grid columns, and hardware-accelerated transforms. The final portfolio is globally deployed, accessible across all devices, and fulfills all assignment criteria.

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